Esteban Alexander Muñoz Cáceres

Electronics Engineer (Ingeniero Civil Electrónico) · Industrial Automation · Software

Chile · eamuzc@gmail.com · +56 9 6261 8697 · linkedin.com/in/eamuz

PROFILE

Electronics engineer (UTEM, Chile) with hands-on experience in automation and maintenance of packaging lines in a dairy plant, PLC programming (Siemens, Allen-Bradley, Delta) and self-driven software work: computer vision, embedded systems and web applications. Available for full-time roles and freelance projects.

EXPERIENCE

2024 – present

Software Development (R&D) · Personal and freelance projects

- Computer vision in Python/OpenCV/YOLOv8, embedded assistants on Raspberry Pi with IoT, and full-stack web apps with React and Node.js.

2023 – present

Automation and maintenance · Soprole — Dairy Plant

- Preventive and corrective maintenance of Tetra Pak Speed A3 packaging lines and packing equipment, planned maintenance (TPM) and diagnostics on Allen-Bradley and Siemens PLCs, following LOTO safety procedures.
- Daily line checks, timely spare-parts requests and maintenance plans for the critical points of each machine.
- Started at the plant as an instrumentation intern (Jan – May 2023): calibration of temperature, flow, level and conductivity instruments in pasteurization, homogenization, fermentation and packaging; STL (AWL) programming on Siemens.

2022

HDPE commissioning project · Dipolitec

- Commissioning of machinery that makes high-density polyethylene pipe: Delta PLC programs for extrusion, cutting, cooling and packing.

2017 – 2022

Electrical installations technician · Indsol and Dipolitec

- Installation and maintenance of electrical, photovoltaic and security systems in residential, commercial and industrial settings; drawing interpretation, testing, measurements and structural welding.
- Photovoltaic and security systems installer (Indsol, 2018 – 2020), junior electrician (Dipolitec, 2020 – 2022) and junior welder (Indsol, 2017 – 2018).

SELECTED PROJECTS

2023

Optimization and Maintenance of a Tetra Pak Speed A3 Packaging Line

Diagnosis and tuning of a high-speed continuous dairy line, eliminating micro-stops at final packaging.

Allen-Bradley ControlLogix · Studio 5000 · Servomotors · TPMS

2022

End-to-End Automation of an HDPE Pipe Extrusion and Cutting Line

Synchronized extrusion and automatic cutting of HDPE pipes, eliminating thermal drift and out-of-sync cuts.

PLC Delta · VFD drives · Encoders · J/K thermocouples

2026

Web SCADA for an HTST Pasteurizer with Roles, Alarms and an Audit Trail

Mimic, temperature loop, flow diversion valve, alarms and trends, with a chained audit trail where every action, allowed or denied, is recorded according to the user role.

SCADA web · Roles and permissions · Chained audit trail · JavaScript

2026

PID Temperature Loop Tuning with Anti-Windup, Simulated on a Pasteurizer

Interactive simulator that compares four tuning methods (Ziegler-Nichols PI and PID, Cohen-Coon and lambda) on a first-order-plus-dead-time plant, with model error, noise and actuator saturation.

Discrete PID · FOPDT · Anti-windup · JavaScript

2026

People Detection, Occupancy and Alarm Platform with Computer Vision

Counts people crossing virtual lines, watches restricted zones with an alarm and photo evidence, and delivers telemetry in Excel and CSV from USB or RTSP cameras.

Python · FastAPI · YOLO (Ultralytics) · OpenCV

2026

AXIOM 2: Voice Desktop Assistant with a Local, Gemini or Claude Brain
Wakes with a double clap, transcribes on the GPU and answers with voice and an animated orb; it controls Windows, files, browser and ComfyUI, and asks for confirmation on risky actions.
Python · Qwen (llama.cpp) · Gemini · Claude

EDUCATION

2015 – 2024

Degree in Civil Electronics Engineering (Ingeniero Civil Electrónico)
Universidad Tecnológica Metropolitana (UTEM), Chile

2022

Diploma in PLC Automation and Programming
Edutin Academy (USA) · Siemens and Allen-Bradley, TIA Portal and RSLogix 5000; Ladder, ST, FBD and SFC

2010 – 2014

Technical secondary education, Electrotechnics program
Internado Nacional Barros Arana, Chile

SKILLS

PLCs & SCADA: Allen-Bradley ControlLogix (RSLogix / Studio 5000) · Siemens S7-300 / TIA Portal · Delta PLC and HMI · FactoryTalk · Ladder, FBD, STL, Structured Text and SFC · SCADA and systems integration · Modbus TCP and device-to-device communication

Instrumentation and control: 4-20 mA / HART · RTD PT100 and J/K thermocouples · Flow, level and conductivity · Pneumatic and modulating valves · PID loops, tuning and anti-windup · Alarm management (ISA-18.2) and OEE · Calibration and commissioning · Dairy processes: pasteurization and packaging

Maintenance & Safety: Preventive and corrective maintenance · Autonomous TPM · LOTO procedures · Online diagnostics · Predictive maintenance (vibration, FFT) · Maintenance plans and spare parts · Operational reliability · Electrical codes and industrial safety

Programming: Python · C / C++ · JavaScript · MATLAB · Visual Basic · PIC assembly · VHDL / Verilog · Object-oriented programming

Computer vision and AI: OpenCV · YOLO (Ultralytics) · PyTorch · Tracking, counting and heatmaps · OCR · Local LLMs (llama.cpp, Qwen) and APIs (Gemini, Claude) · Speech: offline (Vosk) and GPU recognition · Homography, tracking and safety zones

Web, data and deployment: React · Node.js · Django 5 and HTMX · FastAPI · SQLite and databases · Tailwind CSS, WebGL and Canvas 2D · Payments and e-invoicing (Webpay, Flow, OpenFactura) · Automated tests and installers (.exe) · Docker, GitHub and Tailscale (remote access) · Telegram bots

Electronics, embedded and IoT: ESP32 and Arduino (C++) · Raspberry Pi · Microcontroller firmware · IoT with MQTT and home automation (Home Assistant) · PIR / RF sensors and audio (DFPlayer) · Analog and digital circuit analysis · Power electronics · Digital systems (FPGA) and networks

CAD Design & Simulation: AutoCAD (2D and 3D) · Proteus · Eagle · SPICE · Multisim · MATLAB: modeling and numerical simulation · P&ID drawings · Panel and cable sizing · Kinematic synchronization · Cisco Packet Tracer · Cofaso

Generative AI and video: ComfyUI (image and video) · Qwen-Image · Z-Image · Video pipelines with Remotion and ffmpeg · Synthetic voice (Chatterbox) and AI music (ACE-Step) · Reproducible logging: seed, model and license · Local GPU (RTX) for open models

LANGUAGES

Spanish: native · English: intermediate (B1)